

```
1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("traffic_cleaned.csv")

df['date'] = pd.to_datetime(df['date'])

print("Dataset loaded successfully!")
print("Rows:", len(df))
print("Columns:", len(df.columns))
```

Dataset loaded successfully!

Rows: 122567

Columns: 9

```
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import numpy as np
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print("Dataset loaded successfully!")
print("Rows:", len(df))
print("Columns:", len(df.columns))
```

Dataset loaded successfully!

Rows: 122567

Columns: 9

```
]: total_events = len(df)

print("Total Events:", total_events)
```

Total Events: 122567

```
]: event_counts = df['event'].value_counts()

print(event_counts)
```

```
event
pageview    73360
click       32499
preview     16708
Name: count, dtype: int64
```

```
]:
```

```
]: total_events = len(df)

print("Total Events:", total_events)
```

Total Events: 122567

```
]: event_counts = df['event'].value_counts()

print(event_counts)
```

```
event
pageview    73360
click       32499
preview     16708
Name: count, dtype: int64
```

```
]:
```

```
[4]: event_percentage = (  
      df['event'].value_counts(normalize=True) * 100  
    ).round(2)  
    |  
    print(event_percentage)
```

```
event  
pageview    59.85  
click       26.52  
preview     13.63  
Name: proportion, dtype: float64
```

```
[5]: event_summary = pd.DataFrame({  
      'Event': event_counts.index,  
      'Count': event_counts.values,  
      'Percentage': event_percentage.values  
    })  
  
event_summary
```

```
[5]:
```

	Event	Count	Percentage
0	pageview	73360	59.85
1	click	32499	26.52
2	preview	16708	13.63

```
[4]: event_percentage = (  
      df['event'].value_counts(normalize=True) * 100  
    ).round(2)  
    |  
    print(event_percentage)
```

```
event  
pageview    59.85  
click       26.52  
preview     13.63  
Name: proportion, dtype: float64
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```
[5]: event_summary = pd.DataFrame({  
      'Event': event_counts.index,  
      'Count': event_counts.values,  
      'Percentage': event_percentage.values  
    })  
  
event_summary
```

```
[5]:
```

	Event	Count	Percentage
0	pageview	73360	59.85
1	click	32499	26.52
2	preview	16708	13.63

[6]:

```
pageviews = event_counts.get('pageview', 0)
clicks = event_counts.get('click', 0)

pageview_to_click_rate = (clicks / pageviews) * 100

print("Pageview-to-Click Rate:",
      round(pageview_to_click_rate, 2), "%")
```

Pageview-to-Click Rate: 44.3 %

[]:

[7]:

```
previews = event_counts.get('preview', 0)

preview_rate = (previews / pageviews) * 100

print("Preview-to-Pageview Event Rate:",
      round(preview_rate, 2), "%")
```

Preview-to-Pageview Event Rate: 22.78 %

[8]:

[8]:

```
previews = event_counts.get('preview', 0)

preview_rate = (previews / pageviews) * 100
|
print("Preview-to-Pageview Event Rate:",
      round(preview_rate, 2), "%")
```

Preview-to-Pageview Event Rate: 22.78 %

[]:

```
daily_traffic = df.groupby('date').size()
```

```
print(daily_traffic)
```

```
date
2021-08-19    21156
2021-08-20    18536
2021-08-21    16701
2021-08-22    16927
2021-08-23    16415
2021-08-24    16291
2021-08-25    16541
dtype: int64
```

```
plt.figure(figsize=(6, 2.5))
```

```
plt.plot(  
    daily_traffic.index,  
    daily_traffic.values,  
    marker='o'  
)
```

```
plt.title('Daily Website Traffic')  
plt.xlabel('Date')  
plt.ylabel('Total Events')  
plt.xticks(rotation=45)  
plt.grid(alpha=0.3)
```

```
plt.tight_layout()  
plt.show()
```



```
daily_event = (  
    df.groupby(['date', 'event'])  
        .size()  
        .unstack(fill_value=0)  
)  
  
daily_event
```

	event	click	pageview	preview
	date			

2021-08-19	5860	13006	2290
------------	------	-------	------

2021-08-20	4999	11088	2449
------------	------	-------	------

2021-08-21	4279	9939	2483
------------	------	------	------

2021-08-22	4421	9982	2524
------------	------	------	------

2021-08-23	4314	9722	2379
------------	------	------	------

2021-08-24	4231	9705	2355
------------	------	------	------

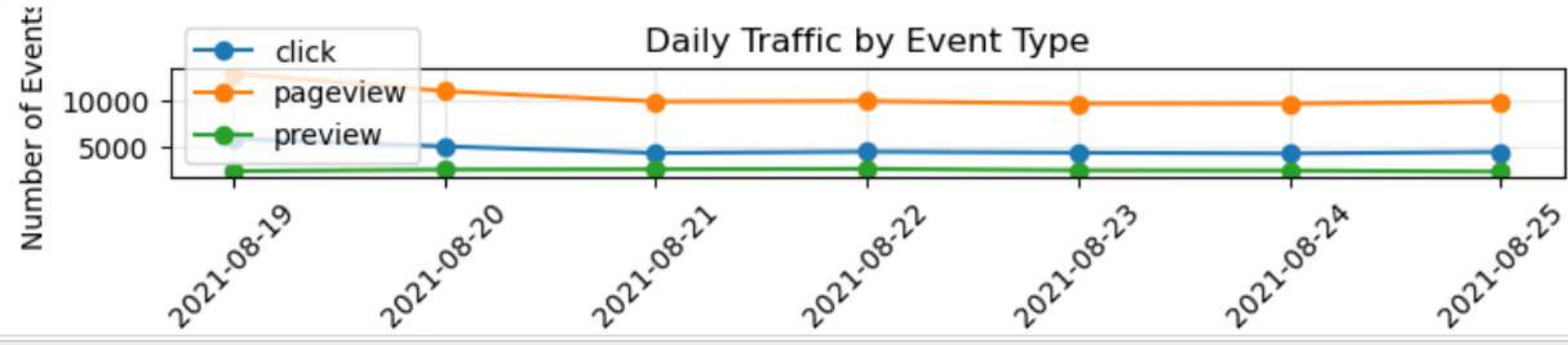
2021-08-25	4395	9918	2228
------------	------	------	------

```
plt.figure(figsize=(8, 2))

for column in daily_event.columns:
    plt.plot(
        daily_event.index,
        daily_event[column],
        marker='o',
        label=column
    )

plt.title('Daily Traffic by Event Type')
plt.xlabel('Date')
plt.ylabel('Number of Events')
plt.xticks(rotation=45)
plt.legend()

plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



```
daily_event = (  
    df.groupby(['date', 'event'])  
        .size()  
        .unstack(fill_value=0)  
)  
  
daily_event|
```

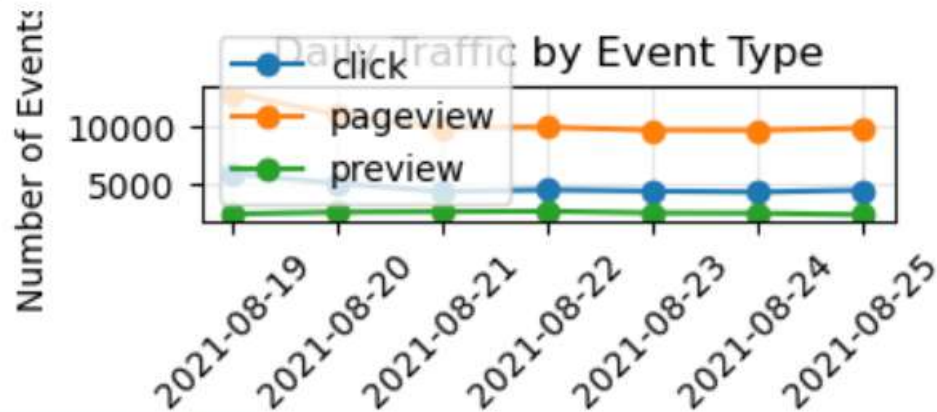
	event	click	pageview	preview
	date			
	2021-08-19	5860	13006	2290
	2021-08-20	4999	11088	2449
	2021-08-21	4279	9939	2483
	2021-08-22	4421	9982	2524
	2021-08-23	4314	9722	2379
	2021-08-24	4231	9705	2355
	2021-08-25	4395	9918	2228

```
plt.figure(figsize=(4,2))

for column in daily_event.columns:
    plt.plot(
        daily_event.index,
        daily_event[column],
        marker='o',
        label=column
    )

plt.title('Daily Traffic by Event Type')
plt.xlabel('Date')
plt.ylabel('Number of Events')
plt.xticks(rotation=45)
plt.legend()

plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



```
: daily_rates = daily_event.div(  
    daily_event.sum(axis=1),  
    axis=0  
) * 100  
  
daily_rates.round(2)
```

```
:  
    event click pageview preview  
    date
```

2021-08-19	27.70	61.48	10.82
2021-08-20	26.97	59.82	13.21
2021-08-21	25.62	59.51	14.87
2021-08-22	26.12	58.97	14.91
2021-08-23	26.28	59.23	14.49
2021-08-24	25.97	59.57	14.46
2021-08-25	26.57	59.96	13.47

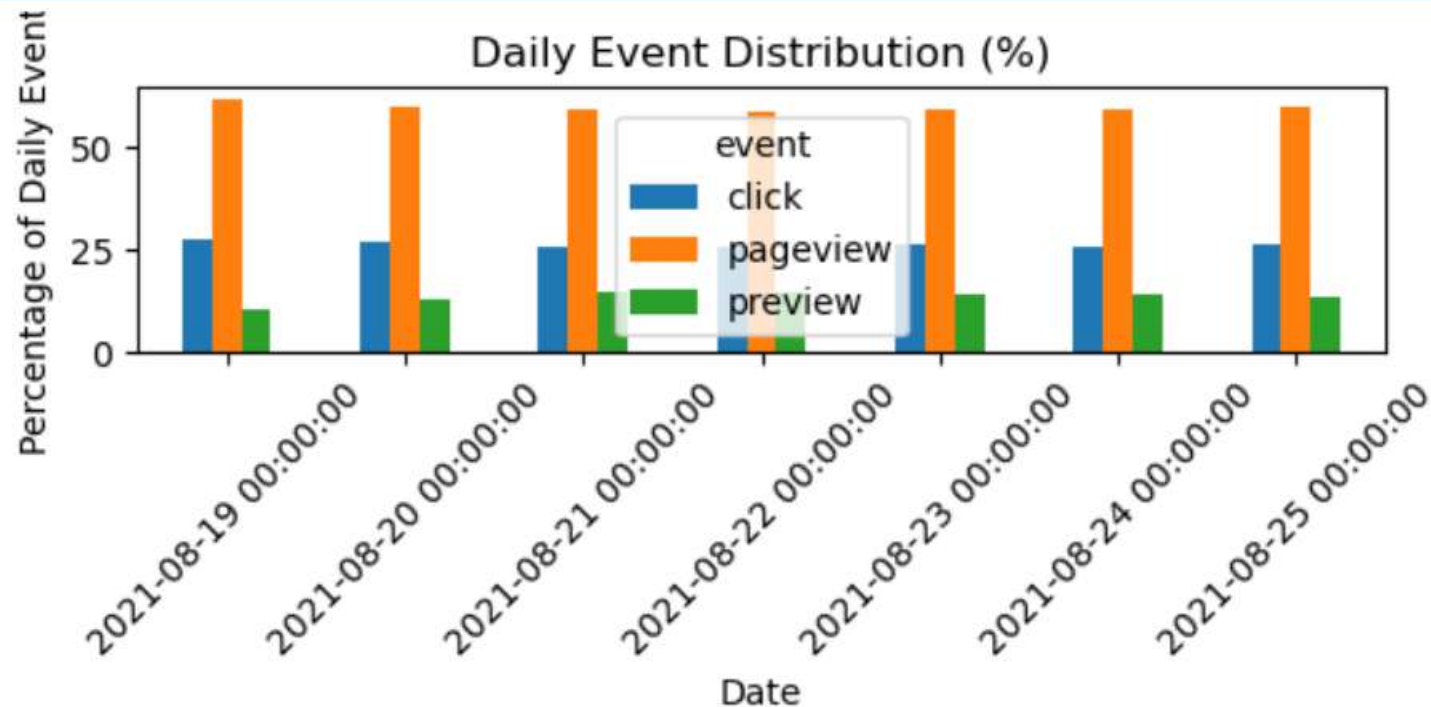
```

daily_rates.plot(
    kind='bar',
    figsize=(6, 3)
)

plt.title('Daily Event Distribution (%)')
plt.xlabel('Date')
plt.ylabel('Percentage of Daily Events')
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()

```



```
country_traffic = (  
    df.groupby('country')  
        .size()  
        .sort_values(ascending=False)  
)  
  
country_traffic.head(10)
```

country	
United States	28664
India	18689
France	10565
Saudi Arabia	7682
United Kingdom	5095
Germany	4017
Canada	2784
Pakistan	2633
Iraq	2444
Turkey	2399

dtype: int64

```
top_10_countries = country_traffic.head(10)
```

```
top_10_countries|
```

country	
United States	28664
India	18689
France	10565
Saudi Arabia	7682
United Kingdom	5095
Germany	4017
Canada	2784
Pakistan	2633
Iraq	2444
Turkey	2399

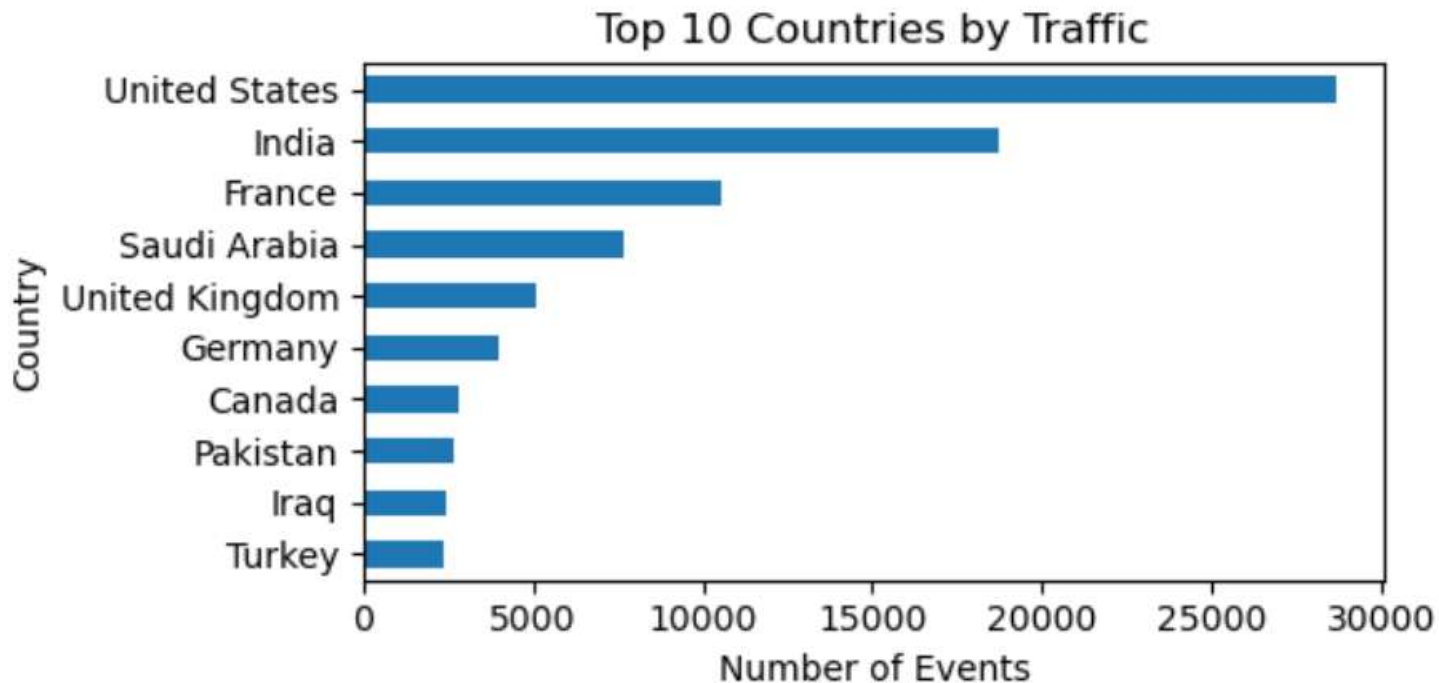
dtype: int64

```
plt.figure(figsize=(6, 3))

top_10_countries.sort_values().plot(kind='barh')

plt.title('Top 10 Countries by Traffic')
plt.xlabel('Number of Events')
plt.ylabel('Country')

plt.tight_layout()
plt.show()
```



```
country_event = pd.crosstab(  
    df['country'],  
    df['event']  
)  
  
country_event.head()
```

event **click** **pageview** **preview**

country

Afghanistan	0	2	0
--------------------	---	---	---

Albania	58	231	58
----------------	----	-----	----

Algeria	96	328	91
----------------	----	-----	----

American Samoa	1	3	0
-----------------------	---	---	---

Andorra	0	2	1
----------------	---	---	---

```
top_countries = country_traffic.head(10).index

country_event_top10 = country_event.loc[top_countries]

country_event_top10
```

	event	click	pageview	preview
country				
United States		8446	18131	2087
India		5360	10565	2764
France		2792	6181	1592
Saudi Arabia		1976	3712	1994
United Kingdom		1307	3218	570
Germany		1030	2585	402
Canada		840	1702	242
Pakistan		754	1382	497
Iraq		567	1285	592
Turkey		466	1598	335

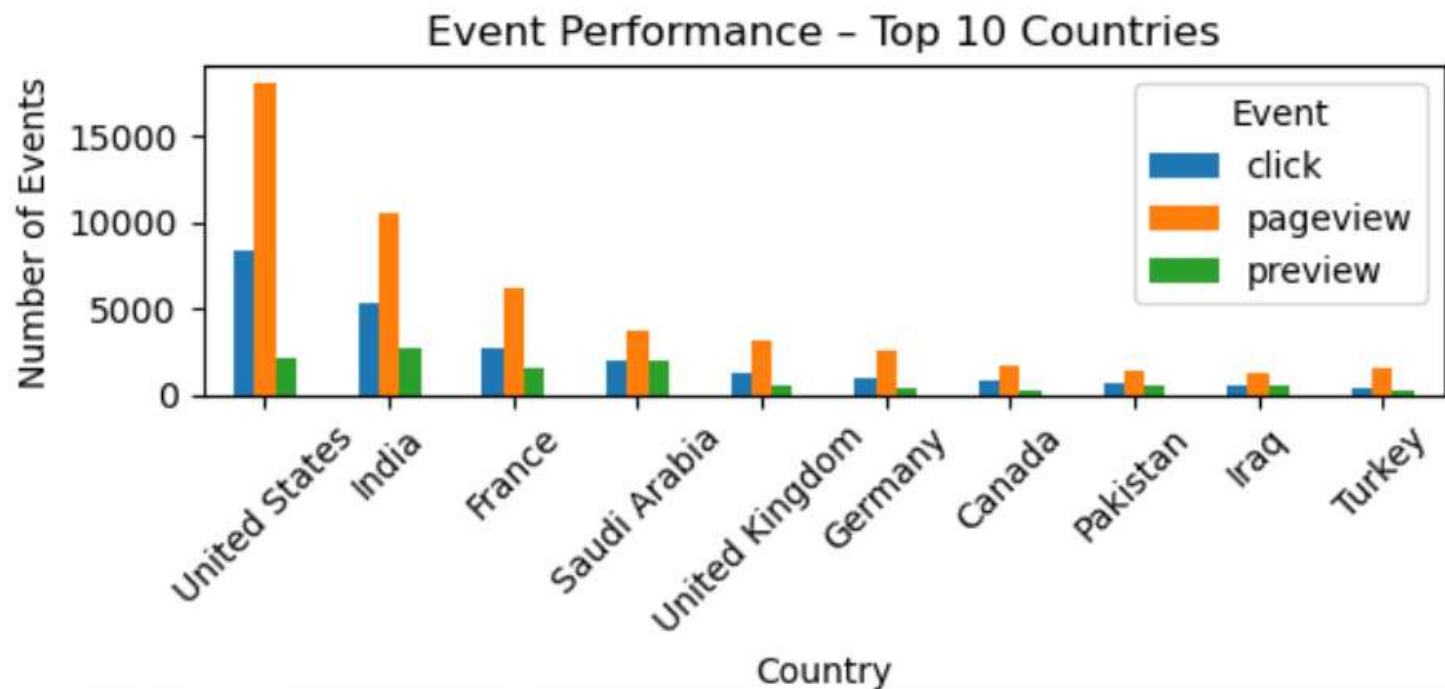
```

]: country_event_top10.plot(
    kind='bar',
    figsize=(6, 3)
)

plt.title('Event Performance - Top 10 Countries')
plt.xlabel('Country')
plt.ylabel('Number of Events')
plt.xticks(rotation=45)
plt.legend(title='Event')

plt.tight_layout()
plt.show()

```



```
country_metrics = pd.crosstab(  
    df['country'],  
    df['event']  
)  
  
country_metrics['click_rate'] = (  
    country_metrics['click'] /  
    country_metrics['pageview']  
) * 100  
  
country_metrics = country_metrics.sort_values(  
    'click_rate',  
    ascending=False  
)  
  
country_metrics.head(10)
```

	event	click	pageview	preview	click_rate
country					
Central African Republic		1	1	0	100.0
Wallis and Futuna		1	1	0	100.0
Iran		2	2	2	100.0
Eritrea		2	2	1	100.0
Principality of Monaco		1	1	0	100.0
Solomon Islands		1	1	0	100.0
Saint Martin		1	1	0	100.0
Guernsey		1	1	0	100.0
Seychelles		2	2	1	100.0
Namibia		12	15	3	80.0

```
country_filtered = country_metrics[
    country_metrics['pageview'] >= 100
]

country_filtered.sort_values(
    'click_rate',|
    ascending=False
).head(10)
```

	event	click	pageview	preview	click_rate
country					
	Ghana	191	311	95	61.414791
	Nigeria	354	636	177	55.660377
	Pakistan	754	1382	497	54.558611
	Saudi Arabia	1976	3712	1994	53.232759
	Mauritania	69	132	39	52.272727
	Netherlands	430	826	173	52.058111
	Kenya	93	180	62	51.666667
	Kuwait	221	433	202	51.039261
	India	5360	10565	2764	50.733554
	Australia	503	998	173	50.400802

```
link_metrics = pd.crosstab(  
    df['linkid'],  
    df['event']  
)  
  
link_metrics['total_events'] = link_metrics.sum(axis=1)  
  
link_metrics = link_metrics.sort_values(  
    'total_events',  
    ascending=False  
)  
  
link_metrics.head(10)
```

:

	event	click	pageview	preview	total_events
	linkid				
	2d896d31-97b6-4869-967b-1c5fb9cd4bb8	2266	4576	1446	8288
	e849515b-929d-44c8-a505-e7622f1827e9	1062	2137	751	3950
	522da5cc-8177-4140-97a7-a84fdb4caf1c	891	2142	753	3786
	a82e72d8-523f-43ab-9fd0-992605d54b4e	908	1736	415	3059
	681d888a-59ce-4acb-b7c5-95dab0c063d9	873	1637	355	2865
	23199824-9cf5-4b98-942a-34965c3b0cc2	684	1559	432	2675
	c2c876ab-b093-4750-9449-6b4913da6af3	579	1391	472	2442
	97105aff-3e40-4c2e-bfeb-563b55e8f5dd	549	1428	413	2390
	d5317591-872a-487d-87cb-10a516643d5f	566	1155	353	2074
	a107c329-5d35-4c7d-8050-50bc69ce7dbc	473	1199	400	2072

```
link_metrics['click_rate'] = (  
    link_metrics['click'] /  
    link_metrics['pageview']  
) * 100
```

```
link_metrics['click_rate'] = (  
    link_metrics['click'] /  
    link_metrics['pageview'].replace(0, np.nan)  
) * 100
```

```
link_metrics[  
    link_metrics['pageview'] >= 20  
>].sort_values(  
    'click_rate',  
    ascending=False  
>).head(10)
```

	event	click	pageview	preview	total_events	click_rate
linkid						
afe8194e-5002-4e56-bc96-060bf355402d	19	24	1	44	79.166667	
ffd3c9e7-c5c5-4f28-b03d-cbaec33f2152	53	67	0	120	79.104478	
47f54e58-af71-4580-9182-364b14b7740a	16	21	3	40	76.190476	
f2cef536-5a20-4dc1-a973-600051bdea04	23	34	3	60	67.647059	
339a61ec-9379-5805-aa3b-baa2f13acb43	19	29	0	48	65.517241	
c7ca19fe-a533-4b8a-85c7-faed28ead5e5	13	20	2	35	65.000000	
08bc2f1c-5c52-4616-acbd-de4e7a62c7bc	25	39	9	73	64.102564	
52762e40-cb7a-4a8a-beb7-03fbd6cca9f8	204	329	66	599	62.006079	
51a9a818-40d2-432d-b48a-9dcff61c6278	13	21	1	35	61.904762	
e1e0b741-3076-405f-a3e9-c613a946c969	13	21	1	35	61.904762	

```
artist_metrics = pd.crosstab(  
    df['artist'],  
    df['event']  
)  
  
artist_metrics['total_events'] = artist_metrics.sum(axis=1)  
  
artist_metrics = artist_metrics.sort_values(  
    'total_events',  
    ascending=False  
)  
  
artist_metrics.head(10)
```

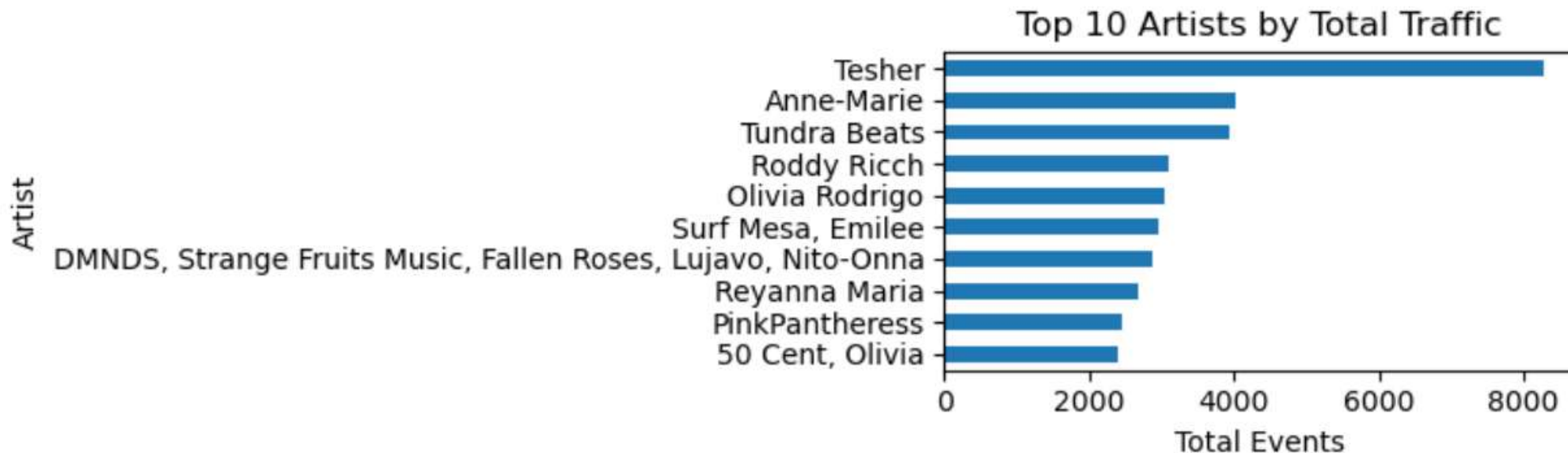
	event	click	pageview	preview	total_events
artist					
Tesher	2266	4576	1446	8288	
Anne-Marie	939	2290	800	4029	
Tundra Beats	1062	2138	751	3951	
Roddy Ricch	928	1763	416	3107	
Olivia Rodrigo	667	1931	439	3037	
Surf Mesa, Emilee	693	1704	559	2956	
DMNDS, Strange Fruits Music, Fallen Roses, Lujavo, Nito-Onna	873	1637	355	2865	
Reyanna Maria	684	1559	432	2675	
PinkPantheress	661	1537	248	2446	
50 Cent, Olivia	549	1428	413	2390	

```
top_artists = artist_metrics.head(10)

top_artists['total_events'].sort_values().plot(
    kind='barh',
    figsize=(8,2.5)
)

plt.title('Top 10 Artists by Total Traffic')
plt.xlabel('Total Events')
plt.ylabel('Artist')

plt.tight_layout()
plt.show()
```



```
: artist_metrics['click_rate'] = (  
    artist_metrics['click'] /  
    artist_metrics['pageview'].replace(0, np.nan)  
) * 100
```

```
: artist_filtered = artist_metrics[  
    artist_metrics['pageview'] >= 50  
]  
  
artist_filtered.sort_values(  
    'click_rate',  
    ascending=False  
)<div data-bbox="33 900 151 955" data-label="Text">

```
).head(10)
```


```

	event	click	pageview	preview	total_events	click_rate
	artist					
	Sahir Ali Bagga	53	67	0	120	79.104478
	Sueco	204	329	66	599	62.006079
Simone & Simaria, Sebastian Yatra		46	77	26	149	59.740260
	Kodak Black	192	329	36	557	58.358663
	Soldier Kidd	214	367	59	640	58.310627
	FINNEAS	51	88	26	165	57.954545
	Lil Tecca, Gunna	277	478	0	755	57.949791
	Nessa Barrett	53	92	12	157	57.608696
Vandelux, Alex Maher		49	86	25	160	56.976744
	Capella Grey	386	678	69	1133	56.932153

```
total_events = len(df)
```

```
pageviews = (df['event'] == 'pageview').sum()
```

```
previews = (df['event'] == 'preview').sum()
```

```
clicks = (df['event'] == 'click').sum()
```

```
pageview_click_rate = clicks / pageviews * 100
```

```
preview_pageview_rate = previews / pageviews * 100
```

```
kpi_summary = pd.DataFrame({  
    'Metric': [  
        'Total Events',  
        'Pageviews',  
        'Previews',  
        'Clicks',  
        'Pageview → Click Event Rate',  
        'Pageview → Preview Event Rate',  
        'Countries',  
        'Cities',  
        'Artists',  
        'Tracks',  
        'Unique Links'
```

```
],
```

```
'Value': [  
    total_events,  
    pageviews,  
    previews,  
    clicks,  
    round(pageview_click_rate, 2),  
    round(preview_pageview_rate, 2),  
    df['country'].nunique(),  
    df['city'].nunique(),  
    df['artist'].nunique(),  
    df['track'].nunique(),  
    df['linkid'].nunique()  
]  
})
```

kpi_summary

[37]:

	Metric	Value
0	Total Events	122567.00
1	Pageviews	73360.00
2	Previews	16708.00
3	Clicks	32499.00
4	Pageview → Click Event Rate	44.30
5	Pageview → Preview Event Rate	22.78
6	Countries	211.00
7	Cities	11993.00
8	Artists	2419.00
9	Tracks	3562.00
10	Unique Links	3839.00

```
kpi_summary.to_csv(  
    'phase2_kpi_summary.csv',  
    index=False  
)
```

```
country_metrics.to_csv(  
    'country_metrics.csv'  
)
```

```
artist_metrics.to_csv(  
    'artist_metrics.csv'  
)
```

```
link_metrics.to_csv(  
    'link_metrics.csv'  
)
```